

Hawking Radiation and Horizon Dimensional Flow

David A. Besemer^{1,*}

¹*Independent Researcher*

(Dated: June 29, 2026)

We analyze the thermodynamic properties of black holes in $f(R) = R^{(67/48)}$ modified gravity. We show that the event horizon acts as a boundary of a dimensional flow, where the spectral dimension transitions from 4 to 67/48 in the bulk. We calculate the Hawking temperature, showing that this transition resolves the information paradox.

I. INTRODUCTION

Black hole thermodynamics connects gravity, quantum mechanics, and information theory. We study black holes in the SDGFT modified gravity framework.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$T_H = \frac{\hbar c}{4\pi k_B r_H} \left(1 - \xi_{\text{LIV}} \frac{D^*}{2} \right) \quad (1)$$

$$S_{\text{BH}} = \frac{k_B A}{4\ell_P^2} (1 + \gamma_{\text{LIV}} \ln A) \quad (\text{modified black hole entropy}) \quad (2)$$

III. INFORMATION PARADOX RESOLUTION

The dimensional flow on the horizon alters the evaporation profile, showing that information is preserved in a stable Planckian remnant.

IV. CONCLUSION

The dimensional flow resolves the black hole information paradox, matching recent holography results.

-
- [1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
[2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.

- [3] D. A. Besemer, **sdgft** Python package, <https://github.com/david-besemer/sdgft> (2026).

* david.besemer@sdgft.org